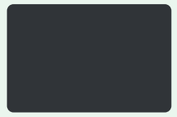
PILZ SAFETY ARCHITECTURE - 11 LOCKED GUARD DOORS

SCHLENKER 3610
VISIBLE POINT-TO-POINT CONNECTIONS

PNOZ m ES

772138
PROFINET

PNOZ m B0

772100
BASE

a4 - EF 16DI

772140
GS1-GS8

a5 - EF 8DI4DO

772142
GS9-GS11

PNOZ m EF 1MM

772170
SPEED

REMAINING SAFETY CONNECTIONS

a5.i6 = physical reset pushbutton

a5.i7 = EDM feedback K1/K2

a1.O0 + a1.O1 = safe stop outputs K1/K2

a4 - 16 safe inputs - GS1 to GS8

a5 - 6 safe inputs - GS9 to GS11

GS1 - FRONT 1 a4	GS2 - FRONT 2 a4	GS3 - FRONT 3 a4	GS4 - FRONT 4 a4	GS5 - REAR 1 a4	GS6 - REAR 2 a4	GS7 - REAR 3 a4	GS8 - REAR 4 a4	GS9 - RIGHT 1 a5	GS10 - RIGHT 2 a5	GS11 - LEFT 1 a5
21-22 a4.i0	21-22 a4.i2	21-22 a4.i4	21-22 a4.i6	21-22 a4.i8	21-22 a4.i10	21-22 a4.i12	21-22 a4.i14	21-22 a5.i0	21-22 a5.i2	21-22 a5.i4
31-32 a4.i1	31-32 a4.i3	31-32 a4.i5	31-32 a4.i7	31-32 a4.i9	31-32 a4.i11	31-32 a4.i13	31-32 a4.i15	31-32 a5.i1	31-32 a5.i3	31-32 a5.i5
E1-E2 KGS1	E1-E2 KGS2	E1-E2 KGS3	E1-E2 KGS4	E1-E2 KGS5	E1-E2 KGS6	E1-E2 KGS7	E1-E2 KGS8	E1-E2 KGS9	E1-E2 KGS10	E1-E2 KGS11

SAFE_UL_24V - individual protected unlock branches

F-UL1 + KGS1 | F-UL2 + KGS2 | F-UL3 + KGS3 | F-UL4 + KGS4 | F-UL5 + KGS5 | F-UL6 + KGS6 | F-UL7 + KGS7 | F-UL8 + KGS8 | F-UL9 + KGS9 | F-UL10 + KGS10 | F-UL11 + KGS11

SAFE UNLOCK PERMISSION

KUL1 + KUL2 force-guided, EDM monitored

Creates SAFE_UL_24V only after safe stop

SAFE_UL_24V →

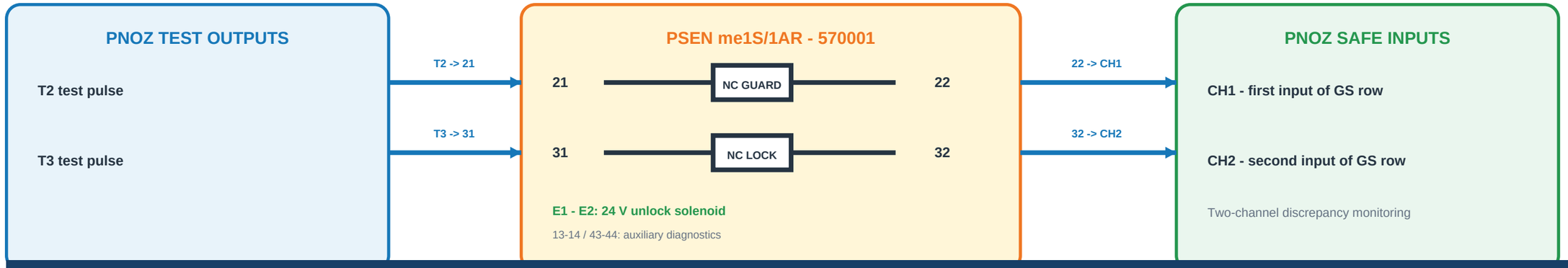
OPERATING SEQUENCE

1 ACCESS REQUEST → 2 CONTROLLED STOP → 3 ZERO SPEED → 4 SAFE UNLOCK → 5 ACCESS → 6 CLOSE + LOCK → 7 RESET → 8 SEPARATE START

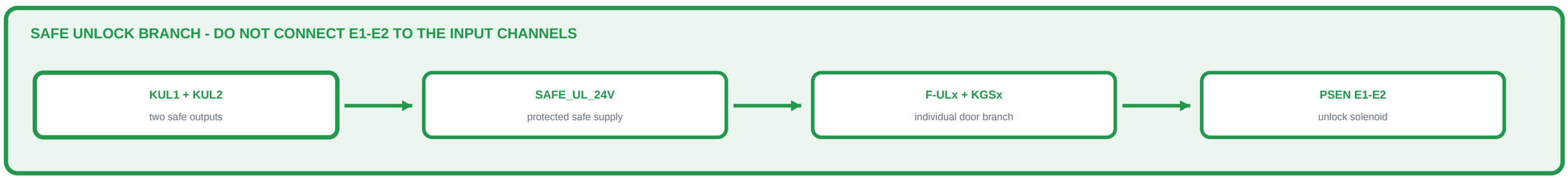
MANDATORY: no automatic restart. Open/unlocked guard, channel discrepancy or EDM fault inhibits restart.

Every branch must be tested individually before commissioning.

TYPICAL CONNECTION - REPEAT FOR GS1 TO GS11



GS / LOCATION	MODULE	CHANNEL 1	CHANNEL 2	UNLOCK	FUNCTION
GS1 FRONT 1	**a4**	21-22 -> a4.i0	31-32 -> a4.i1	E1-E2 / KGS1	guard + lock
GS2 FRONT 2	**a4**	21-22 -> a4.i2	31-32 -> a4.i3	E1-E2 / KGS2	guard + lock
GS3 FRONT 3	**a4**	21-22 -> a4.i4	31-32 -> a4.i5	E1-E2 / KGS3	guard + lock
GS4 FRONT 4	**a4**	21-22 -> a4.i6	31-32 -> a4.i7	E1-E2 / KGS4	guard + lock
GS5 REAR 1	**a4**	21-22 -> a4.i8	31-32 -> a4.i9	E1-E2 / KGS5	guard + lock
GS6 REAR 2	**a4**	21-22 -> a4.i10	31-32 -> a4.i11	E1-E2 / KGS6	guard + lock
GS7 REAR 3	**a4**	21-22 -> a4.i12	31-32 -> a4.i13	E1-E2 / KGS7	guard + lock
GS8 REAR 4	**a4**	21-22 -> a4.i14	31-32 -> a4.i15	E1-E2 / KGS8	guard + lock
GS9 RIGHT 1	**a5**	21-22 -> a5.i0	31-32 -> a5.i1	E1-E2 / KGS9	guard + lock
GS10 RIGHT 2	**a5**	21-22 -> a5.i2	31-32 -> a5.i3	E1-E2 / KGS10	guard + lock
GS11 LEFT 1	**a5**	21-22 -> a5.i4	31-32 -> a5.i5	E1-E2 / KGS11	guard + lock



VALIDATION BEFORE ENERGISING: test each door, both channels, discrepancy, safe stop, unlock permission, EDM and separate restart.

Use this architecture together with the released electrical drawing containing wire numbers, terminal blocks, fuse ratings and cable references.